



Main Title: USA Car Sales Market Analysis

Subtitle: LEVERAGING DATA TO OPTIMIZE INVENTORY AND PRICING STRATEGY

PROJECT OUTLINE

❑ INTRODUCTION

❑ PROJECT OBJECTIVES

❑ DATA CLEANING AND VISUALIZATION (DASHBOARD)

❑ OBSERVATIONS AND RECOMMENDATIONS/INSIGHT

INTRODUCTION

The Landscape of the US Secondary Auto Market

The US automotive secondary market is currently undergoing significant shifts. Changes in consumer brand loyalty, combined with evolving standards for vehicle valuation based on mileage and physical condition, have made traditional "gut feeling" decision-making obsolete.

To stay competitive, businesses must transit to a data-driven model. This analysis leverages a comprehensive dataset of **2,499 vehicle listings** across the United States to provide a definitive look at market performance and profitability drivers.

PROJECT OBJECTIVES

Transforming Data into Actionable Strategy

The primary goal of this project is to provide a roadmap for optimizing sales and sourcing through four key objectives:

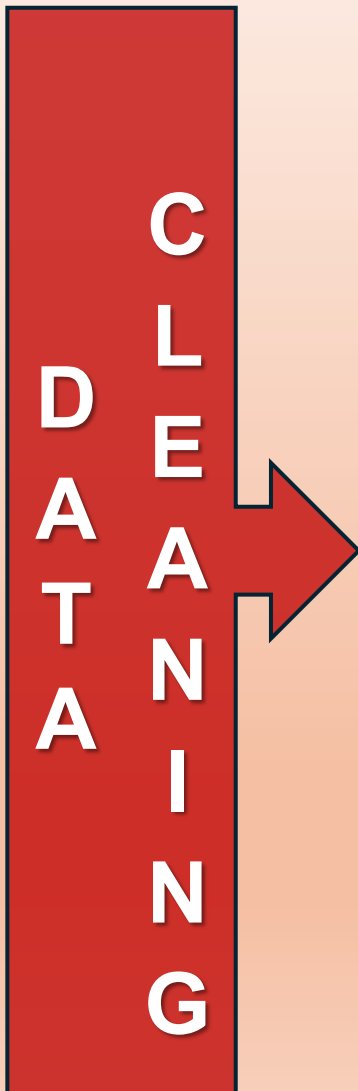
Market Dominance: Pinpoint which brands currently hold the largest market share to prioritize high-volume inventory.

Geographic Optimization: Identify high-revenue states to streamline logistics, shipping, and sourcing efforts.

Risk & Valuation: Quantify the exact financial impact of vehicle titles — specifically comparing the value retention of **Clean vs. Salvage** units.

The "Sweet Spot": To know the ideal intersection where mileage and price offer the maximum value to the customer while maintaining healthy margins.

The Data was sourced from the Kaggle (USA_cars_datasets) in CSV format; Microsoft Excel was used in the analysis for the cleaning and Power BI was used for the visualization.



MY PROJECT PRECIOUS - Excel

File Home Insert Draw Page Layout Formulas Data Review View Help

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1		price	brand	model	year	title_status	mileage	color	vin	lot	state	country	condition	
2	0	6300	toyota	cruiser	2008	clean vehicle	274117	black	jtezu11f88k007763	159348797	new jersey	usa	10 days left	
3	1	2899	ford	se	2011	clean vehicle	190552	silver	2fmdk3gc4bbb02217	166951262	tennessee	usa	6 days left	
4	2	5350	dodge	mpv	2018	clean vehicle	39590	silver	3c4pdcgg5jt346413	167655728	georgia	usa	2 days left	
5	3	25000	ford	door	2014	clean vehicle	64146	blue	1ftfw1et4efc23745	167753855	virginia	usa	22 hours left	
6	4	27700	chevrolet	1500	2018	clean vehicle	6654	red	3gcpcrec2jg473991	167763266	florida	usa	22 hours left	
7	5	5700	dodge	mpv	2018	clean vehicle	45561	white	2c4rdgeg9jr237989	167655771	texas	usa	2 days left	
8	6	7300	chevrolet	pk	2010	clean vehicle	149050	black	1gcsksea1az121133	167753872	georgia	usa	22 hours left	
9	7	13350	gmc	door	2017	clean vehicle	23525	gray	1gks2gkc3hr326762	167692494	california	usa	20 hours left	
10	8	14600	chevrolet	malibu	2018	clean vehicle	9371	silver	1g1zd5st5jf191860	167763267	florida	usa	22 hours left	
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13	11	12920	gmc	mpv	2017	clean vehicle	39650	white	1gks2bkc6hr136280	167692496	california	usa	20 hours left	
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15	13	5430	chrysler	wagon	2017	clean vehicle	138650	gray	2c4rc1cg5hr616095	167656123	texas	usa	2 days left	
16	14	20700	ford	door	2013	clean vehicle	100757	black	1ftfw1et7dfa47790	167753916	virginia	usa	22 hours left	
17	15	12710	gmc	door	2017	clean vehicle	25747	white	1gks2gkc6hr328389	167692497	california	usa	20 hours left	
18	16	5200	kia	forte	2018	clean vehicle	46194	blue	3kpfk4a77je198723	167801757	north carolina	usa	2 days left	
19	17	16500	buick	encore	2018	clean vehicle	20002	red	kl4cj1sb6jb688844	167763275	tennessee	usa	22 hours left	
20	18	5210	ford	mpv	2017	clean vehicle	35714	white	2fmpk3j95hbb73607	167656124	texas	usa	2 days left	
21	19	38100	ford	door	2013	clean vehicle	54380	gray	1ft8w3dt5deb68569	167753923	virginia	usa	22 hours left	
22	20	12520	gmc	door	2017	clean vehicle	30114	white	1gks2gkcxhr328380	167692498	california	usa	20 hours left	
23	21	7300	kia	forte	2018	clean vehicle	38823	black	3kpf14a79je272611	167801773	north carolina	usa	2 days left	
24	22	15000	chevrolet	door	2015	clean vehicle	61578	black	2gnflhe33f6214023	167763280	ohio	usa	2 days left	
25	23	6180	ford	mpv	2017	clean vehicle	46010	blue	2fmpk3j84hbc09276	167656125	texas	usa	2 days left	
26	24	20800	ford	door	2013	clean vehicle	105510	silver	1ftfw1efxdfa01008	167753937	virginia	usa	22 hours left	

DIRTY DATA CLEANED DATA PIVOT TABLE Drop Down

DATA CLEANING

	C	D	E	F	G	H	I	J	K	L	M	N	O	P
3	Customer ID	Price	Brand	Model	Year	Title Status	Mileage	Color	VIN	Lot	State	Country	Condition	
4	1	6300	Toyota	Cruiser	2008	Clean Vehicle	274117	Black	jtezu11f88k007763	159348797	New Jersey	USA	10 days left	
5	2	2899	Ford	Se	2011	Clean Vehicle	190552	Silver	2fmdk3gc4bbb02217	166951262	Tennessee	USA	6 days left	
6	3	5350	Dodge	Mpv	2018	Clean Vehicle	39590	Silver	3c4pdcgg5jt346413	167655728	Georgia	USA	2 days left	
7	4	25000	Ford	Door	2014	Clean Vehicle	64146	Blue	1ftfw1et4efc23745	167753855	Virginia	USA	22 hours left	
8	5	27700	Chevrolet	1500	2018	Clean Vehicle	6654	Red	3gcpcrec2jg473991	167763266	Florida	USA	22 hours left	
9	6	5700	Dodge	Mpv	2018	Clean Vehicle	45561	White	2c4rdgeg9jr237989	167655771	Texas	USA	2 days left	
10	7	7300	Chevrolet	Pk	2010	Clean Vehicle	149050	Black	1gcsksea1az121133	167753872	Georgia	USA	22 hours left	
11	8	13350	Gmc	Door	2017	Clean Vehicle	23525	Gray	1gks2gkc3hr326762	167692494	California	USA	20 hours left	
12	9	14600	Chevrolet	Malibu	2018	Clean Vehicle	9371	Silver	1g1zd5st5jf191860	167763267	Florida	USA	22 hours left	
13	10	5250	Ford	Mpv	2017	Clean Vehicle	63418	Black	2fmpk3j92hbc12542	167656121	Texas	USA	2 days left	
14	11	10400	Dodge	Coupe	2009	Clean Vehicle	107856	Orange	2b3lj54t49h509675	167753874	Georgia	USA	22 hours left	
15	12	12920	Gmc	Mpv	2017	Clean Vehicle	39650	White	1gks2bkc6hr136280	167692496	California	USA	20 hours left	
16	13	31900	Chevrolet	1500	2018	Clean Vehicle	22909	Black	3gcukrec0jg176059	167763273	Tennessee	USA	22 hours left	
17	14	5430	Chrysler	Wagon	2017	Clean Vehicle	138650	Gray	2c4rc1cg5hr616095	167656123	Texas	USA	2 days left	
18	15	20700	Ford	Door	2013	Clean Vehicle	100757	Black	1ftfw1et7dfa47790	167753916	Virginia	USA	22 hours left	
19	16	12710	Gmc	Door	2017	Clean Vehicle	25747	White	1gks2gkc6hr328389	167692497	California	USA	20 hours left	
20	17	5200	Kia	Forte	2018	Clean Vehicle	46194	Blue	3kpfk4a77je198723	167801757	North Carolina	USA	2 days left	
21	18	16500	Buick	Encore	2018	Clean Vehicle	20002	Red	kl4cj1sb6jb688844	167763275	Tennessee	USA	22 hours left	
22	19	5210	Ford	Mpv	2017	Clean Vehicle	35714	White	2fmpk3j95hbb73607	167656124	Texas	USA	2 days left	
23	20	38100	Ford	Door	2013	Clean Vehicle	54380	Gray	1ft8w3dt5deb68569	167753923	Virginia	USA	22 hours left	
24	21	12520	Gmc	Door	2017	Clean Vehicle	30114	White	1gks2gkcxhr328380	167692498	California	USA	20 hours left	
25	22	7300	Kia	Forte	2018	Clean Vehicle	38823	Black	3kpf14a79je272611	167801773	North Carolina	USA	2 days left	
26	23	15000	Chevrolet	Door	2015	Clean Vehicle	61578	Black	2gnflhe33f6214023	167763280	Ohio	USA	2 days left	
27	24	6180	Ford	Mpv	2017	Clean Vehicle	46010	Blue	2fmpk3j84hbc09276	167656125	Texas	USA	2 days left	



VIN

All

USA CAR SALES MARKET ANALYSIS

52.3K

Average of Mileage

\$18.8K

Average of Price...

\$46.9M

Sum of Price (USD)

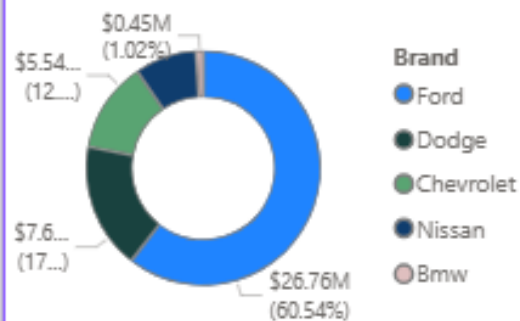
2499

Cars Sold

28

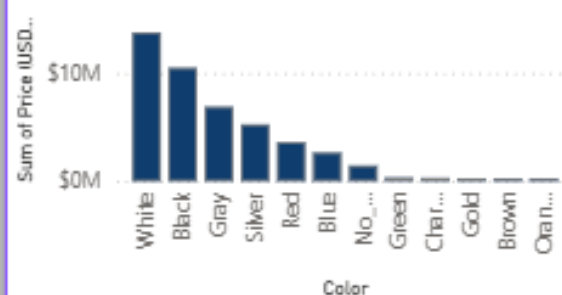
Count of Brand

Sales by Brand



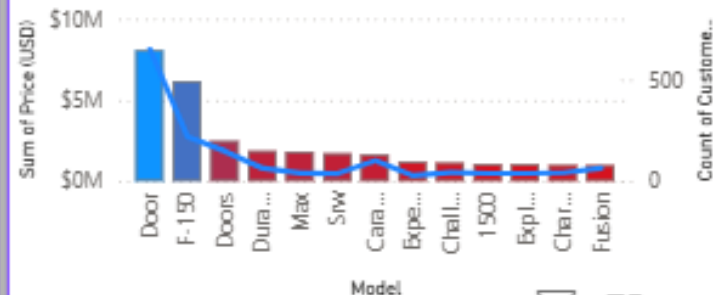
Sales by Color

Sum of Price (USD) Count of Model

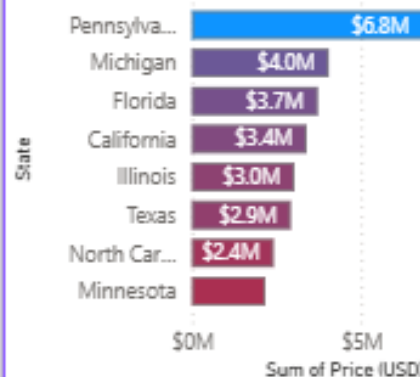


Sales by Model

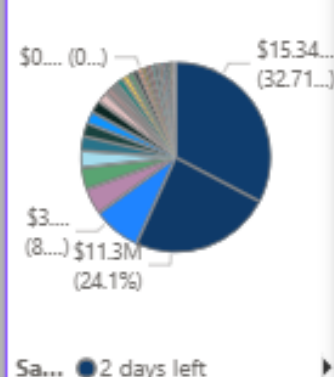
Sum of Price (USD) Count of Customer ID



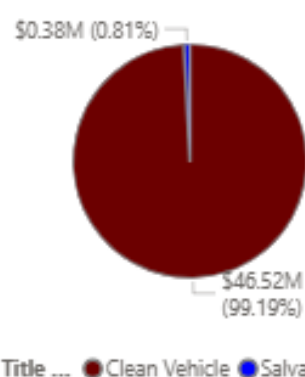
Sales by State



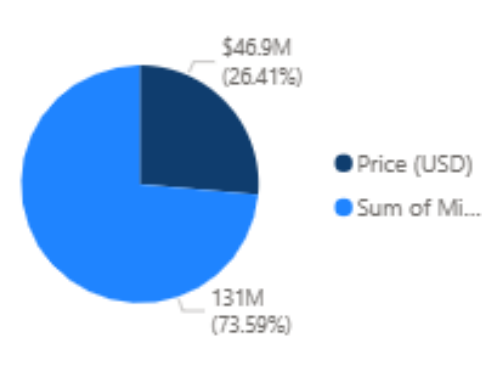
Sales by Condition



Sales by Status



Price (USD) and Sum of Mileage



Color

All

Model

All

Brand

All

Customer ID

All

State

All

Title Status

All

Country

All

Year

All



SALES MARKET ANALYSIS



THE BIG PICTURE

- Total Revenue: \$46.9 Million
- Total Units Sold: 2,499 Vehicles
- Average Selling Price: \$18,767
 - Top Market: Pennsylvania & Michigan (Revenue Leaders)
- Dominant Brand: Ford (Nearly 50% of the dataset)

BRAND PERFORMANCE & VOLUME

Key Insights:

Ford is the primary driver of revenue.

Dodge and **Chevrolet** provide steady secondary volume.

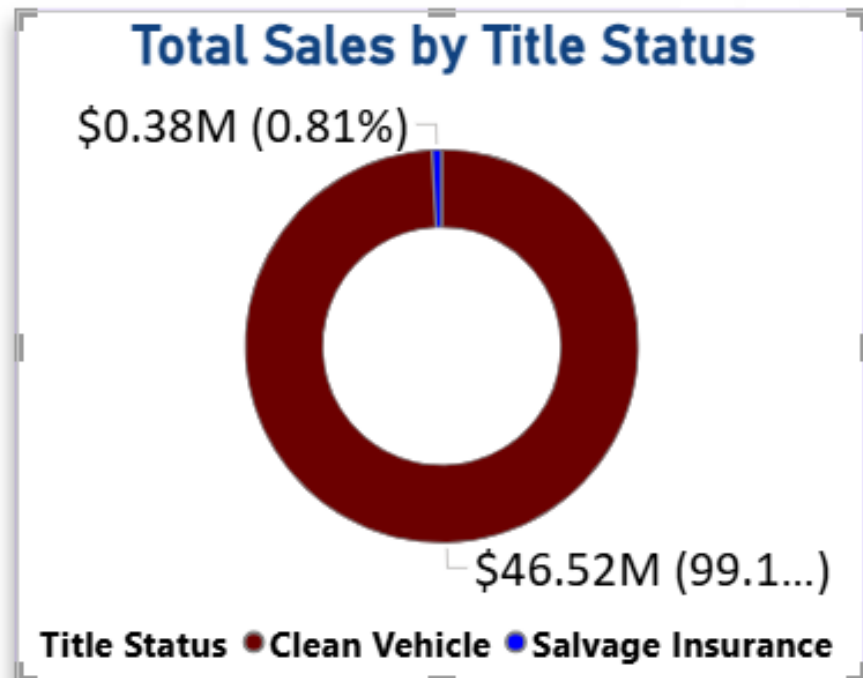
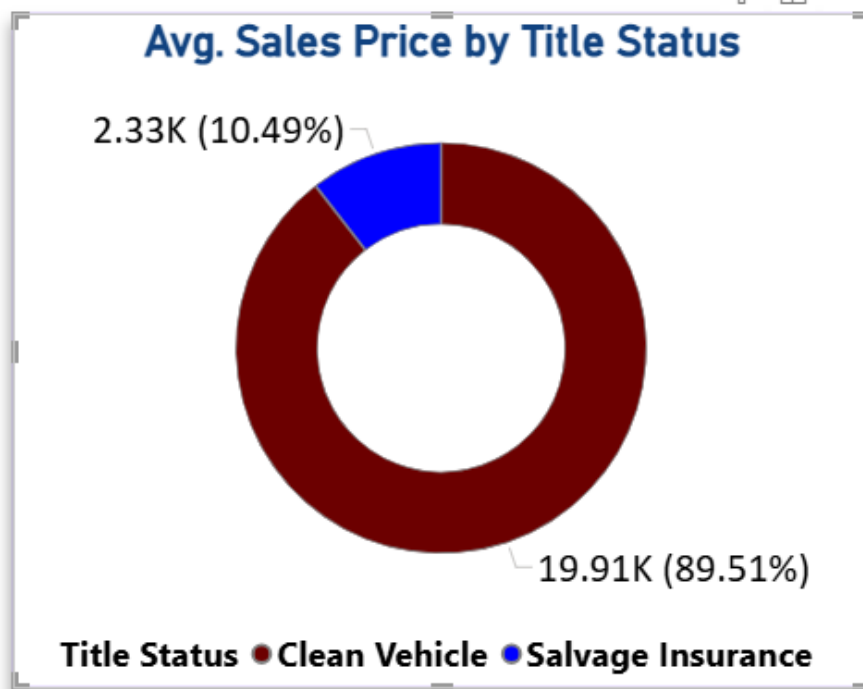
Niche luxury brands like **BMW** and **Mercedes-Benz** command the highest per-unit prices.

THE "CLEAN TITLE" PREMIUM

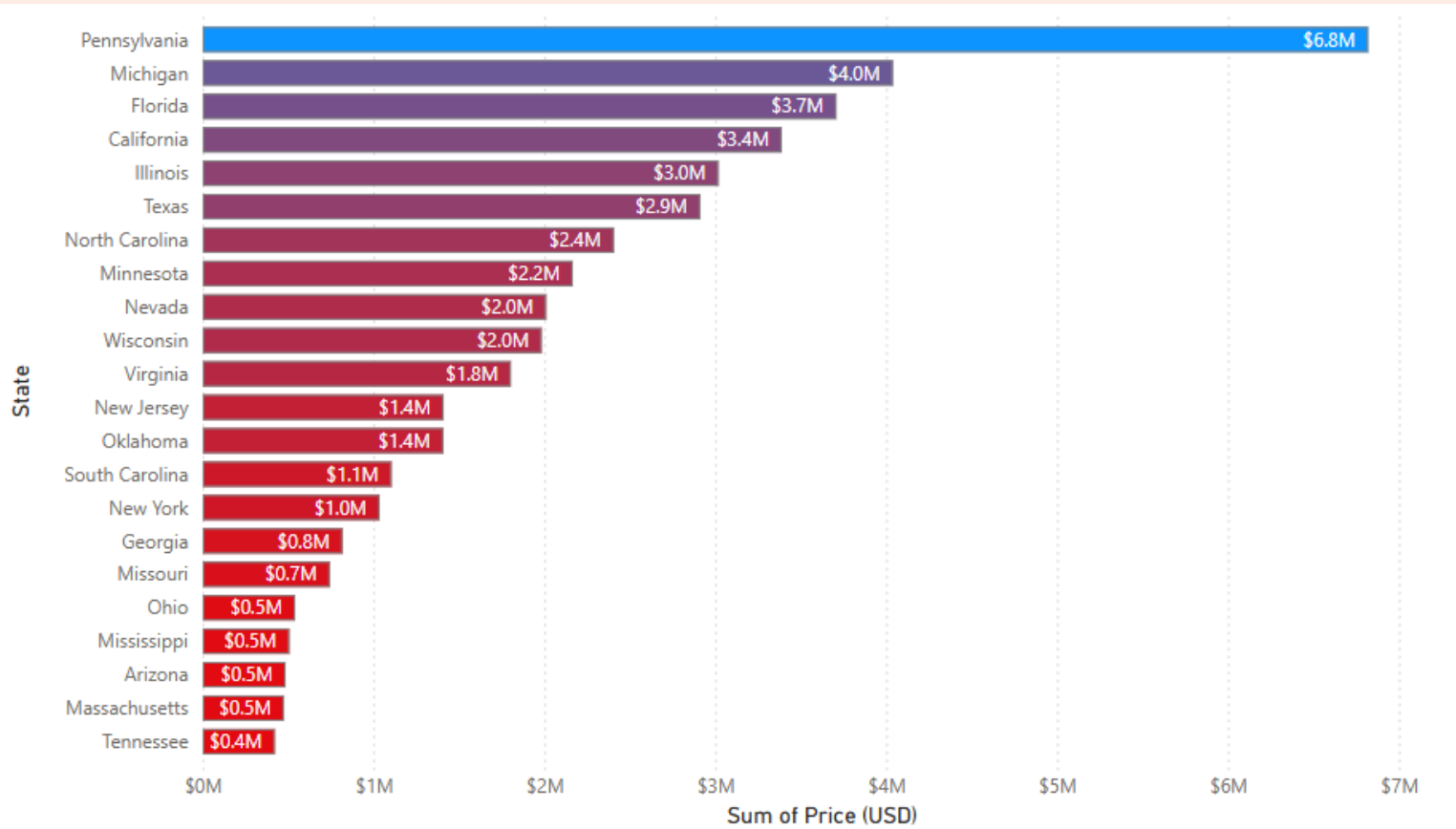
The Discovery:

- ❖ "Clean" vehicles sell for an average of **\$19,914**.
- ❖ "Salvage" vehicles sell for an average of **\$2,333**.

Conclusion: Maintaining a "Clean" title status is the single most important factor in preserving vehicle value.



REGIONAL SALES DISTRIBUTION



- **Visual Suggestion:** A Map of the USA colored by sales volume.

- **Key Insights:**

- **Pennsylvania** leads in transaction count.
- **Michigan** offers the highest average sale price.

Texas and Florida are high-volume but show lower average prices, suggesting a more competitive or budget-friendly market.

OBSERVATIONS AND RECOMMENDATIONS

1. Inventory & Pricing Strategy (The "Title Status" Gap)

Observation: There is a massive price disparity between vehicles with a "**Clean Vehicle**" title (Avg. **\$19,914**) and those with "**Salvage Insurance**" (Avg. **\$2,333**).

Recommendation: For dealer looking for high-margin flipping, the "Salvage" market is cheap, but for high-revenue stability, I recommend prioritizing "Clean Vehicle" titles. The data shows that a "Clean" title adds roughly **8.5x more value** to a car in the dataset.

2. Focus on "Sweet Spot" Brands

Premium Segment: Mercedes-Benz (\$28.7k) and **BMW (\$26.4k)** command the highest average prices. To maximize revenue per sale, focus on these brands.

Volume Leader: Ford is your most important brand, accounting for nearly 50% of the listings (1,235 cars) with a strong average price of **\$21,666**.

Best Value (Low Mileage): Buick represents a hidden gem in the dataset. It has a high average price (\$19.7k) but significantly lower average mileage (**~37.9k miles**) compared to Ford (~52k miles). This suggests Buicks hold their value well with less wear.

3. Regional Targeting

- **Expansion Opportunity:** Pennsylvania is top-performing state by volume (299 listings) and has a high average sale price of **\$22,810**.
- **High-Value Market:** Michigan actually has a higher average sale price (**\$23,899**) than Pennsylvania, despite fewer listings.
- **Recommendation:** Marketing efforts and inventory sourcing should be concentrated in the **Northeast (PA, MI)** where cars are selling at a premium compared to Texas (\$13.6k) or Florida (\$15.0k).

4. "Economy" Options for Quick Turnaround

- **Observation:** Brands like **Hyundai (\$5.2k)** and **Honda (\$6.1k)** have the lowest entry prices but high mileage (Honda averages 91k miles).
- **Recommendation:** These brands are ideal for "budget-conscious" or "first-time buyer" segments. They offer high liquidity (they sell fast because they are cheap) even if the profit per unit is lower.

5. STRATEGIC RECOMMENDATIONS

- High-Margin Focus:** Increase inventory of Mercedes-Benz and BMW in the Michigan market.

- Volume Stability:** Maintain a consistent stock of Ford and Chevrolet to ensure steady cash flow.

- Market Sourcing:** Shift sourcing efforts toward Pennsylvania to capitalize on high buyer activity.

INSIGHT

First, optimize inventory by focusing on the 'Northeast Premium'—bringing more luxury stock to Michigan and Pennsylvania.

Use Ford as liquidity anchor to keep cash flowing.

we should explicitly market our low-mileage Buicks as 'Best Value' alternatives to the more expensive premium brands."

Summary Table for Decision Making

Category	Brand / State	Why?
Highest Revenue	Ford	High volume (1,235 units) and solid pricing.
Premium Choice	Mercedes-Benz	Highest avg. price (\$28,704).
Best Regional Market	Pennsylvania	Most active market with above-average prices.
Budget/Liquidity	Hyundai	Lowest average price (\$5,203) for quick sales.
Quality/Value	Buick	Lower mileage than average for a sub-\$20k price.